

Robot Dynamics & Control

2026

Tutors:

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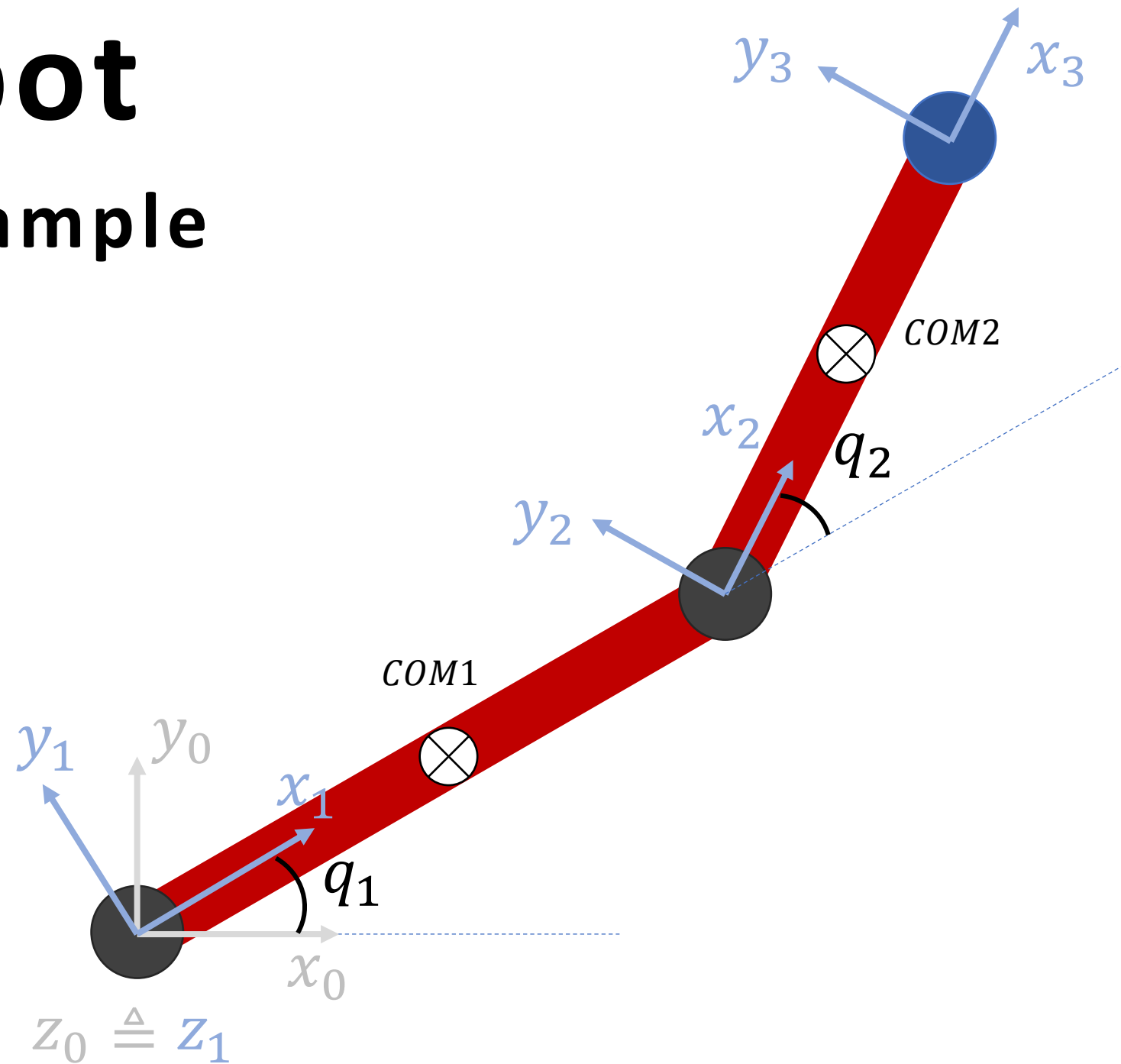
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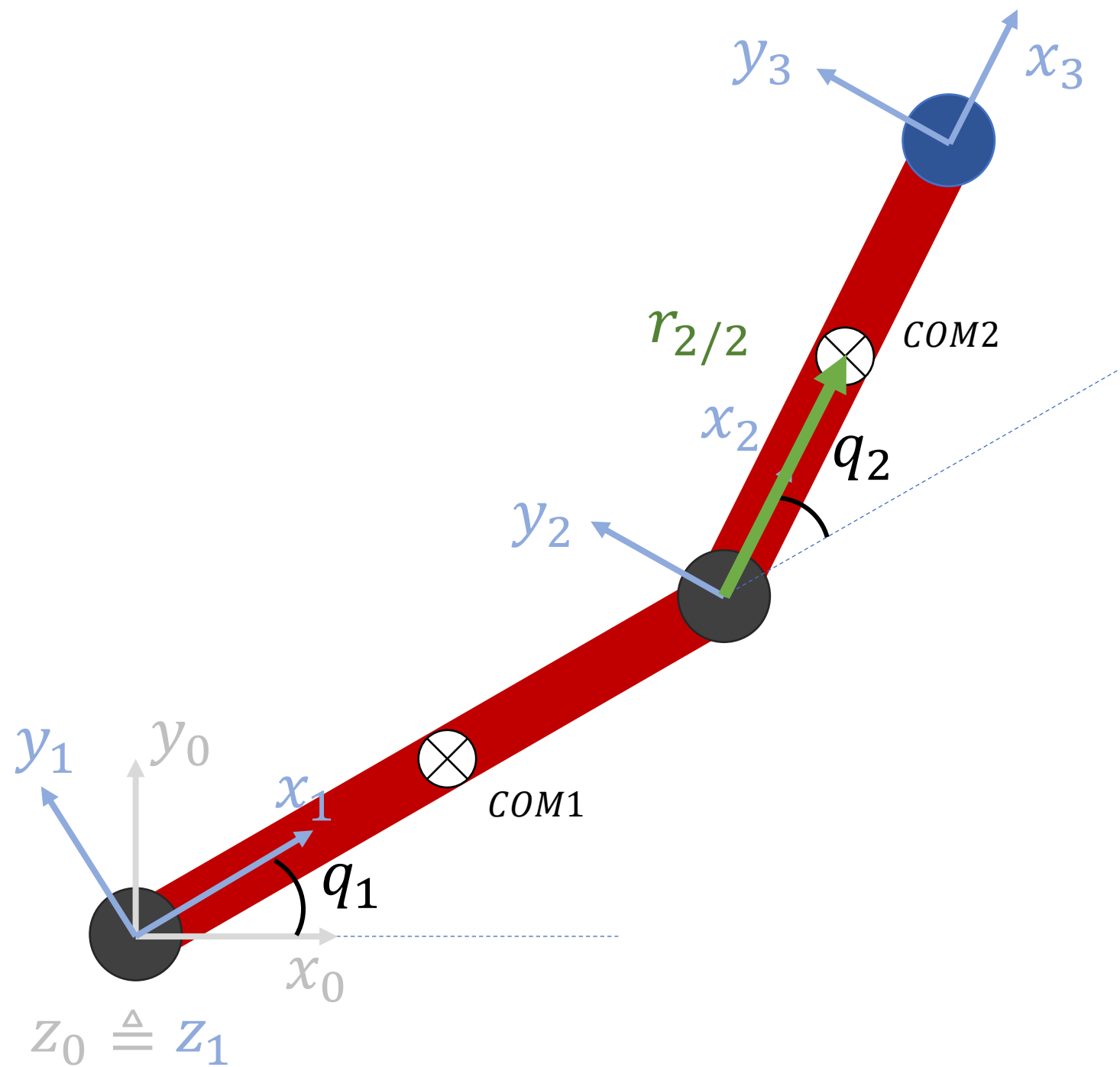
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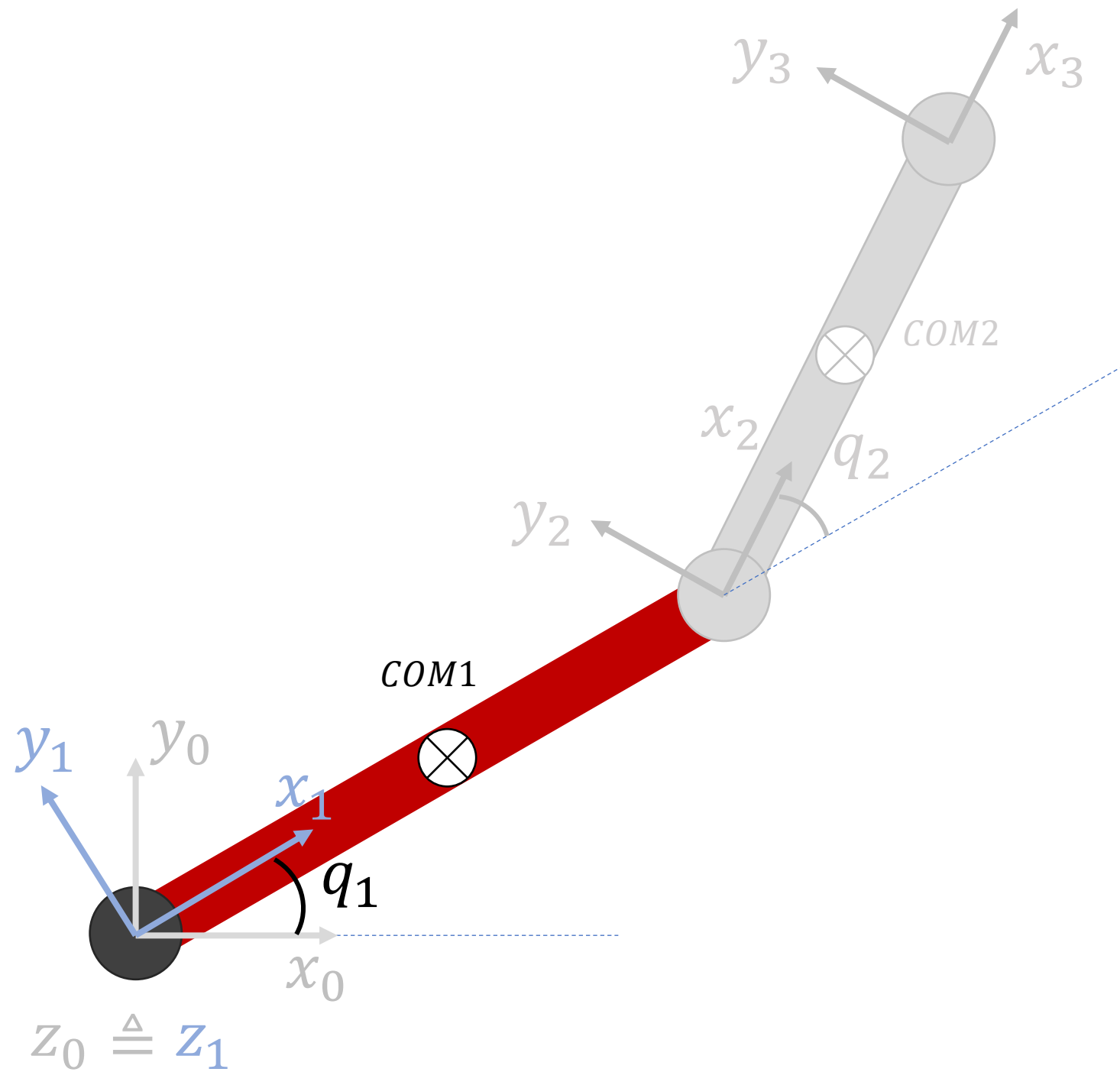


2R Robot

Jacobian Example







Analytic Jacobian computation:

[More intuitive than the geometric Jacobian computation but the result is the same]

First of all, we need to compute the Positions of Centers of Mass:

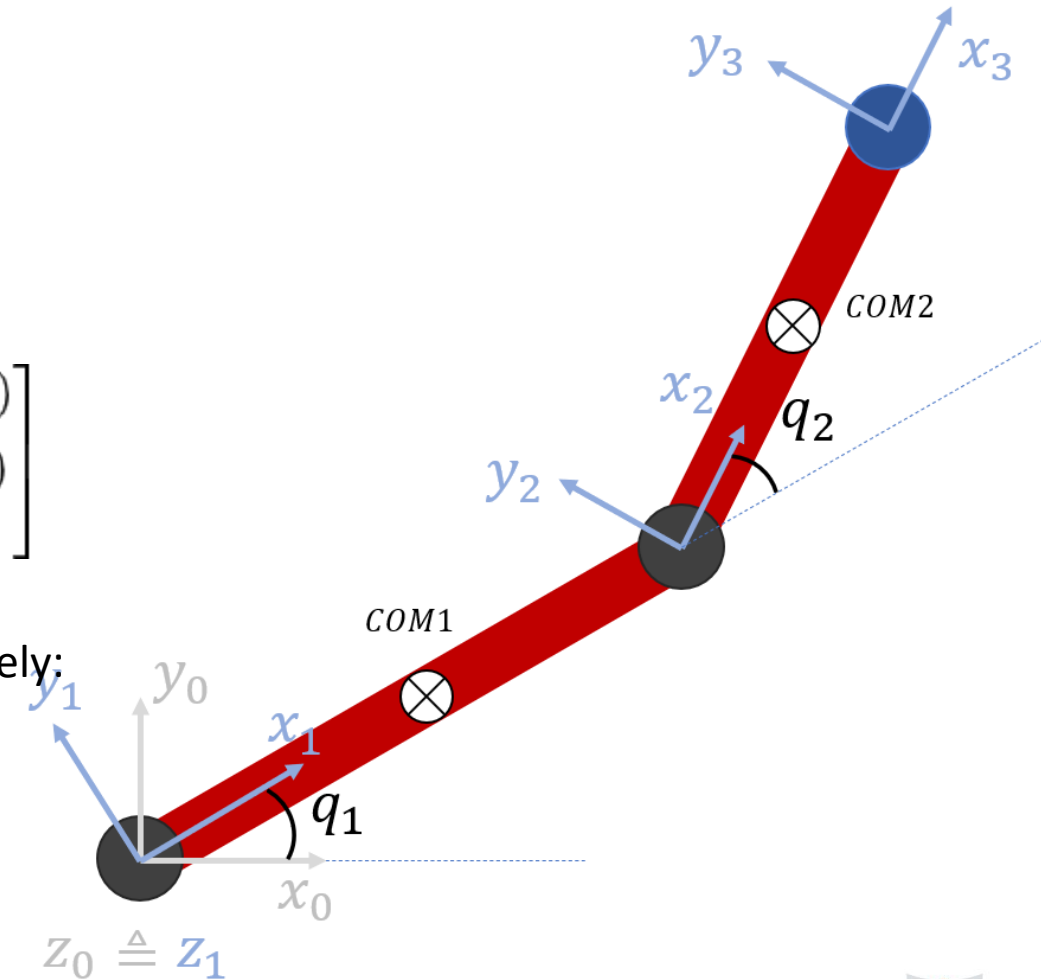
$${}^1x_{c_1} = \begin{bmatrix} \frac{l_1}{2} \\ 0 \\ 0 \end{bmatrix} \quad {}^2x_{c_2} = \begin{bmatrix} \frac{l_2}{2} \\ 0 \\ 0 \end{bmatrix}$$

The Positions of Centers of Mass in the world frame are respectively:

$${}^0x_{c_1} = {}^0R(q_1){}^1x_{c_1} = \begin{bmatrix} \frac{l_1}{2}\cos(q_1) \\ \frac{l_1}{2}\sin(q_1) \\ 0 \end{bmatrix} \quad {}^0x_{c_2} = \begin{bmatrix} l_1\cos(q_1) + \frac{l_2}{2}\cos(q_1 + q_2) \\ l_1\sin(q_1) + \frac{l_2}{2}\sin(q_1 + q_2) \\ 0 \end{bmatrix}$$

The Linear Velocities of Centers of Mass in the world frame are respectively:

$${}^0\dot{x}_{c_1} = \begin{bmatrix} -\frac{l_1}{2}\sin(q_1)\dot{q}_1 \\ \frac{l_1}{2}\cos(q_1)\dot{q}_1 \\ 0 \end{bmatrix} \quad {}^0\dot{x}_{c_2} = \begin{bmatrix} -l_1\sin(q_1)\dot{q}_1 - \frac{l_2}{2}\sin(q_1 + q_2)(\dot{q}_1 + \dot{q}_2) \\ l_1\cos(q_1) + \frac{l_2}{2}\cos(q_1 + q_2)(\dot{q}_1 + \dot{q}_2) \\ 0 \end{bmatrix}$$



These 2 quantities, depending on $\dot{q}_1$ and $\dot{q}_2$ can be used to directly compute the Jacobian matrices of interest



The linear component of the Jacobian matrix projected on frame <0> computed for the CoM1 is:

$${}^0J_{1/0}^L = \begin{bmatrix} {}^0J_{1/1}^L & {}^0J_{1/2}^L \end{bmatrix} = \begin{bmatrix} {}^0J_{1/1}^L & 0 \end{bmatrix} \quad \text{where,} \quad {}^0J_{1/1}^L = \begin{bmatrix} -\frac{l_1}{2} \sin(q_1) \\ \frac{l_1}{2} \cos(q_1) \\ 0 \end{bmatrix}$$

$\downarrow$
 $j > i$ $\uparrow$

The linear component of the Jacobian matrix projected on frame <0> computed for the CoM2 is:

$${}^0J_{2/0}^L = \begin{bmatrix} {}^0J_{2/1}^L & {}^0J_{2/2}^L \end{bmatrix} = \begin{bmatrix} -l_1 \sin(q_1) - \frac{l_2}{2} \sin(q_1 + q_2) & -\frac{l_2}{2} \sin(q_1 + q_2) \\ l_1 \cos(q_1) + \frac{l_2}{2} \cos(q_1 + q_2) & \frac{l_2}{2} \cos(q_1 + q_2) \\ 0 & 0 \end{bmatrix}$$

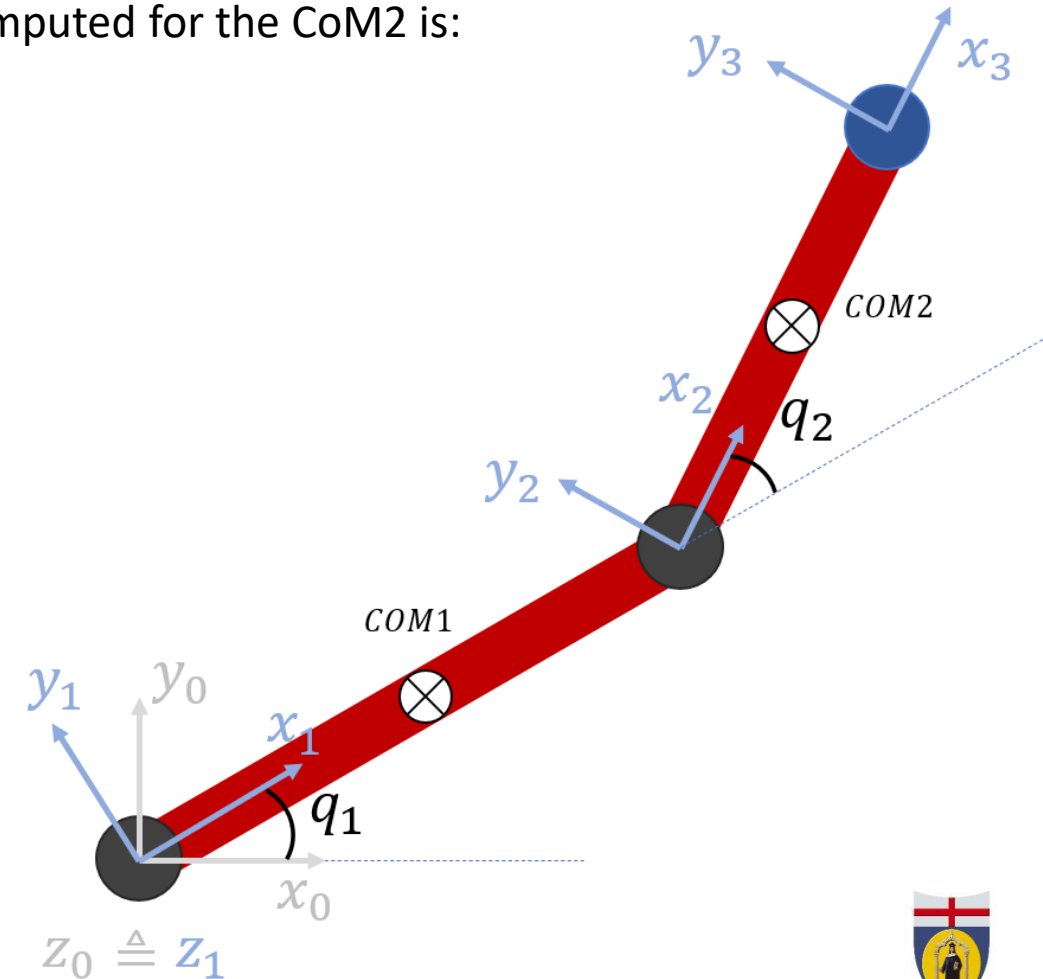
$\underbrace{\hspace{10em}}_{\dot{q}_1} \qquad \underbrace{\hspace{10em}}_{\dot{q}_2}$

The angular component of the Jacobian matrix projected on frame <0> computed for the CoM1 is:

$${}^0J_{1/0}^A = \begin{bmatrix} {}^0J_{1/1}^A & {}^0J_{1/2}^A \end{bmatrix} = \begin{bmatrix} {}^0J_{1/1}^A & 0 \end{bmatrix} \quad \text{where,} \quad {}^0J_{1/1}^A = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The angular component of the Jacobian matrix projected on frame <0> computed for the CoM2 is:

$${}^0J_{2/0}^A = \begin{bmatrix} {}^0J_{2/1}^A & {}^0J_{2/2}^A \end{bmatrix} \quad \text{where,} \quad {}^0J_{2/1}^A = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad {}^0J_{2/2}^A = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



Geometric Jacobian computation:

[Same formulas
on your notes]

The linear component of the Jacobian operator computed for the CoM i with respect to Joint j is:

$$J_{i/j}^L = \begin{cases} (k_j \wedge r_{i/j}) & \text{If } j \text{ is a rotational joint} \\ k_j & \text{If } j \text{ is a prismatic joint} \end{cases} \quad \text{If } j > i \quad J_{i/j}^L = 0$$

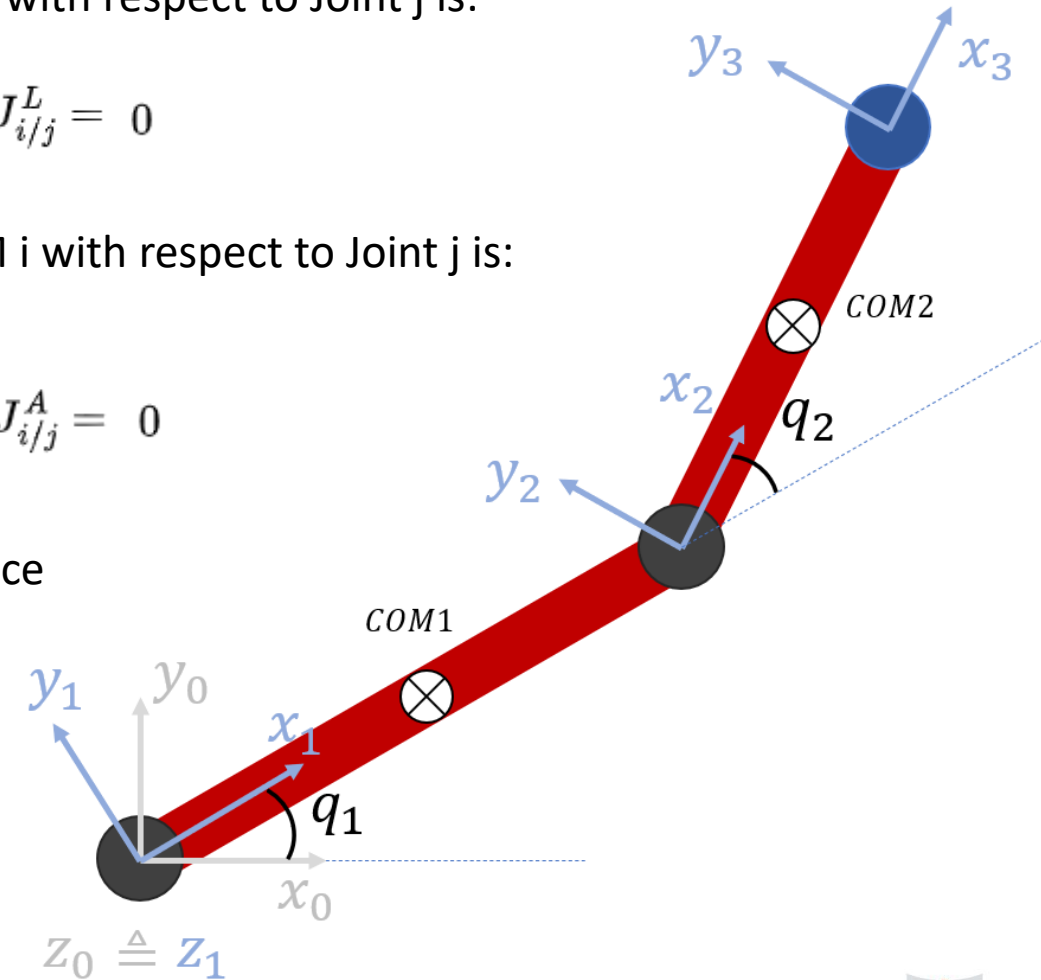
The angular component of the Jacobian operator computed for the CoM i with respect to Joint j is:

$$J_{i/j}^A = \begin{cases} k_j & \text{If } j \text{ is a rotational joint} \\ 0 & \text{If } j \text{ is a prismatic joint} \end{cases} \quad \text{If } j > i \quad J_{i/j}^A = 0$$

To become useful quantities the Jacobian must be projected on a reference frame, e.g. the base frame:

$${}^0 J_{i/0}^L = \begin{bmatrix} {}^0 J_{i/1}^L & {}^0 J_{i/2}^L \end{bmatrix}$$

$${}^0 J_{i/0}^A = \begin{bmatrix} {}^0 J_{i/1}^A & {}^0 J_{i/2}^A \end{bmatrix}$$



Let's consider the CoM1:

The Linear component of the Jacobian matrix is:

$${}^0J_{1/0}^L = \begin{bmatrix} {}^0J_{1/1}^L & {}^0J_{1/2}^L \end{bmatrix} = \begin{bmatrix} {}^0J_{1/1}^L & 0 \end{bmatrix}, \text{ where:}$$

$${}^0J_{1/1}^L = {}^0k_1 \wedge {}^0r_{1/1} \quad \text{such that:}$$

$${}^0k_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad {}^0r_{1/1} = \begin{bmatrix} \frac{l_1}{2} \cos(q_1) \\ \frac{l_1}{2} \sin(q_1) \\ 0 \end{bmatrix} \Rightarrow$$

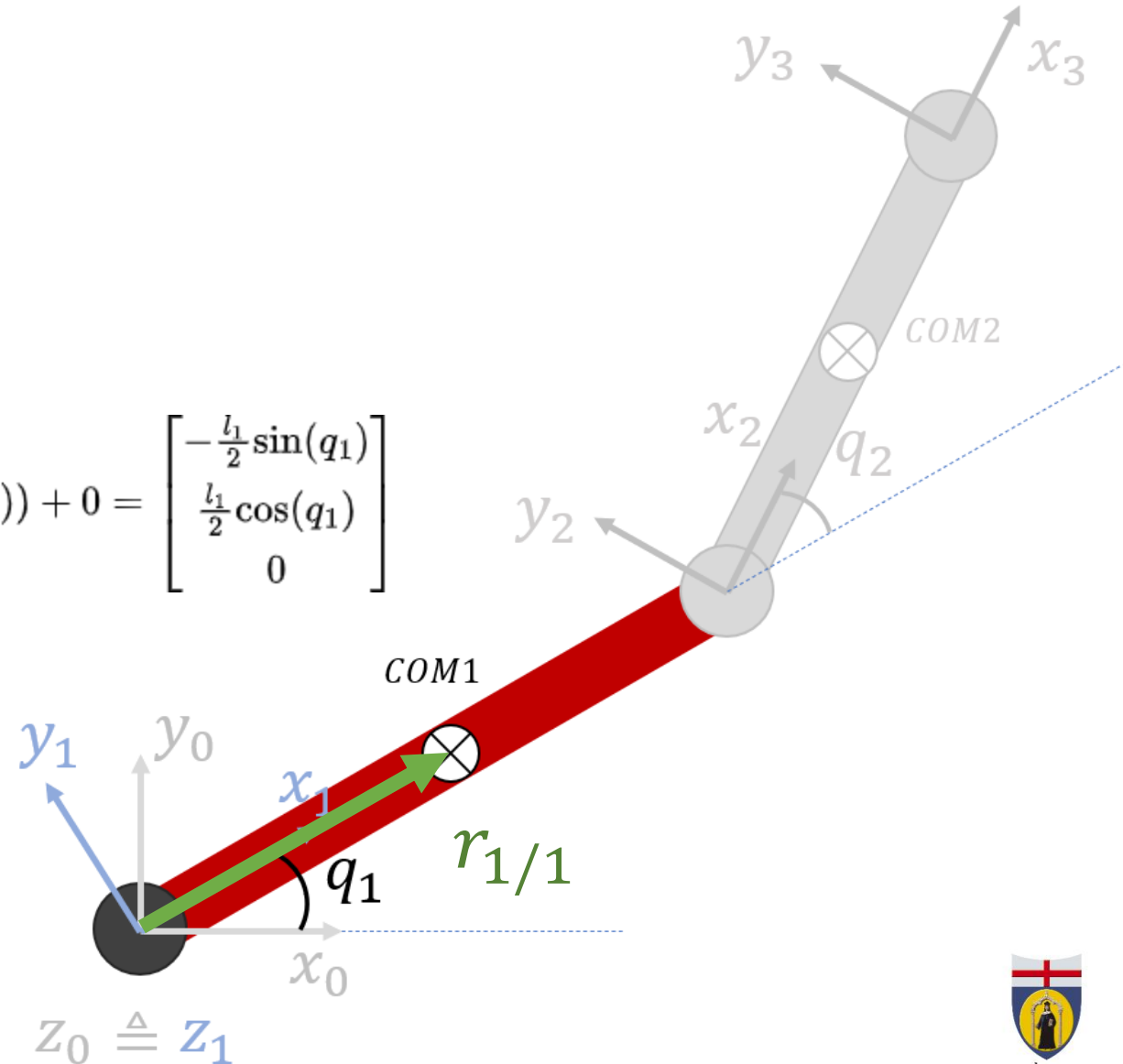
$${}^0J_{1/1}^L = \begin{vmatrix} i & j & k \\ 0 & 0 & 1 \\ \frac{l_1}{2} \cos(q_1) & \frac{l_1}{2} \sin(q_1) & 0 \end{vmatrix} = i\left(-\frac{l_1}{2} \sin(q_1)\right) - j\left(-\frac{l_1}{2} \cos(q_1)\right) + 0 = \begin{bmatrix} -\frac{l_1}{2} \sin(q_1) \\ \frac{l_1}{2} \cos(q_1) \\ 0 \end{bmatrix}$$

The Angular component of the Jacobian matrix is simply:

$${}^0J_{i/1}^A = \begin{bmatrix} {}^0J_{1/1}^A & {}^0J_{2/1}^A \end{bmatrix} = \begin{bmatrix} {}^0J_{1/1}^A & 0 \end{bmatrix} \quad \text{such that:}$$

$${}^0J_{1/1}^A = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

[Same formulas on your notes]



Let's consider the CoM2:

[Same formulas on your notes]

The Linear component of the Jacobian matrix is:

$${}^0J_{2/0}^L = \begin{bmatrix} {}^0J_{2/1}^L & {}^0J_{2/2}^L \end{bmatrix}, \text{ where:}$$

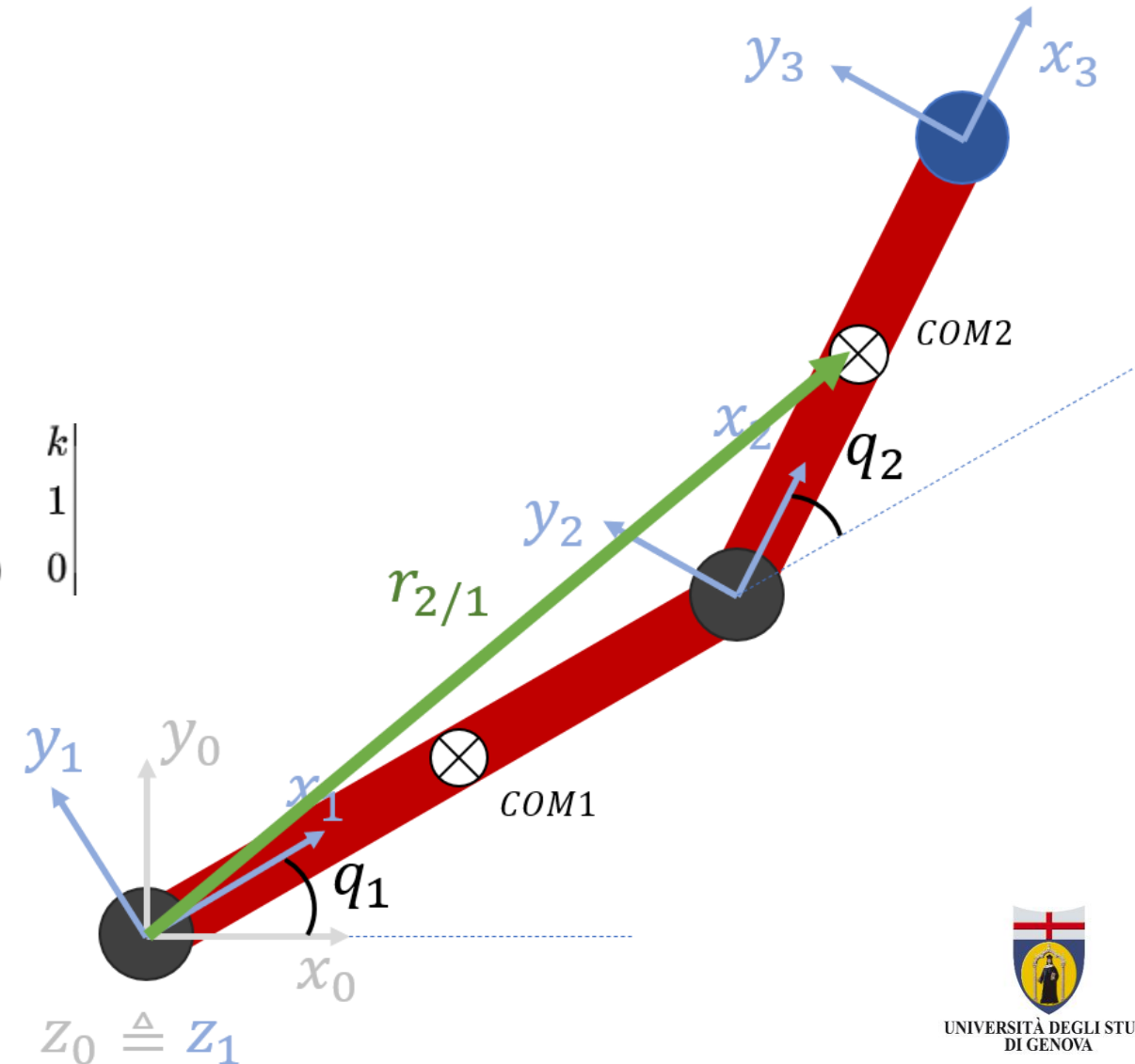
$${}^0J_{2/1}^L = ({}^0k_1 \wedge {}^0r_{2/1}) \quad \text{and} \quad {}^0J_{2/2}^L = ({}^0k_2 \wedge {}^0r_{2/2})$$

Let's consider ${}^0J_{2/1}^L$:

$${}^0k_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad {}^0r_{2/1} = \begin{bmatrix} l_1 \cos(q_1) + \frac{l_2}{2} \cos(q_1 + q_2) \\ l_1 \sin(q_1) + \frac{l_2}{2} \sin(q_1 + q_2) \\ 0 \end{bmatrix}$$

$${}^0J_{2/1}^L = \begin{vmatrix} & i & j & k \\ & 0 & 0 & 1 \\ l_1 \cos(q_1) + \frac{l_2}{2} \cos(q_1 + q_2) & l_1 \sin(q_1) + \frac{l_2}{2} \sin(q_1 + q_2) & 0 \end{vmatrix}$$

$${}^0J_{2/1}^L = \begin{bmatrix} -l_1 \sin(q_1) - \frac{l_2}{2} \sin(q_1 + q_2) \\ l_1 \cos(q_1) + \frac{l_2}{2} \cos(q_1 + q_2) \\ 0 \end{bmatrix}$$



Let's consider ${}^0J_{2/2}^L$:

$${}^0k_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad {}^0r_{2/2} = {}^0r_{2/1} - {}^0P_2 = \begin{bmatrix} l_1 \cos(q_1) + \frac{l_2}{2} \cos(q_1 + q_2) & -l_1 \cos(q_1) \\ l_1 \sin(q_1) + \frac{l_2}{2} \sin(q_1 + q_2) & -l_1 \sin(q_1) \\ 0 & 0 \end{bmatrix} \quad {}^0r_{2/2} = {}^0r_{2/1} - {}^0P_2 = \begin{bmatrix} \frac{l_2}{2} \cos(q_1 + q_2) \\ \frac{l_2}{2} \sin(q_1 + q_2) \\ 0 \end{bmatrix}$$

$${}^0J_{2/2}^L = \begin{vmatrix} i & j & k \\ 0 & 0 & 1 \\ \frac{l_2}{2} \cos(q_1 + q_2) & \frac{l_2}{2} \sin(q_1 + q_2) & 0 \end{vmatrix} = i(-\frac{l_2}{2} \sin(q_1 + q_2)) - j(-\frac{l_2}{2} \cos(q_1 + q_2)) + 0 = \begin{bmatrix} -\frac{l_2}{2} \sin(q_1 + q_2) \\ \frac{l_2}{2} \cos(q_1 + q_2) \\ 0 \end{bmatrix}$$

$${}^0J_{2/0}^L = \begin{bmatrix} {}^0J_{2/1}^L & {}^0J_{2/2}^L \end{bmatrix} = \begin{bmatrix} -l_1 \sin(q_1) - \frac{l_2}{2} \sin(q_1 + q_2) & -\frac{l_2}{2} \sin(q_1 + q_2) \\ l_1 \cos(q_1) + \frac{l_2}{2} \cos(q_1 + q_2) & \frac{l_2}{2} \cos(q_1 + q_2) \\ 0 & 0 \end{bmatrix}$$

